

RESEARCH GUIDE

The Longevity Research Guide

A plain-English breakdown of the compounds serious longevity researchers are actually tracking — what they do, what the evidence shows, and what to ask your doctor before you start.

01	How to Read This Guide	02	Cellular Energy & Repair
03	Weight & Metabolism	04	Recovery & Tissue Repair
05	Hormone Support	06	Longevity & Mitochondria
07	Cognitive Health	08	Skin & Cellular Repair
09	Why Source Quality Matters	10	Physician-Supervised Care

Physician-supervised programs. Transparent pricing.

All compounds in this guide are available through licensed physicians — pre-mixed, compounded to order, shipped direct.

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CHAPTER 01

How to Read This Guide

Longevity medicine moves fast. New compounds enter the conversation every year, clinical data accumulates, and the gap between what researchers are studying and what most people hear about keeps growing.

This guide is my attempt to close that gap — at least for the compounds I think are worth understanding. Each section covers what a compound actually does, what the research shows (including the honest gaps), and what questions to ask if you're considering it.

THREE THINGS TO KEEP IN MIND

- **This is not medical advice.** I'm a researcher, not a physician. Nothing here should replace a conversation with your doctor.
- **Evidence quality varies.** Some compounds have solid human trial data. Others have strong animal models and clinical experience but limited large-scale RCTs. I'll be clear about which is which.
- **Source matters enormously.** The same compound from a licensed compounding pharmacy and from a research chemical supplier are very different things. Chapter 9 covers this in detail.

The compounds in this guide are ones I've spent real time researching. I've tried to give you the honest version — not the hype, not the fear, just what the data actually shows.

— Erica

CHAPTER 02

Cellular Energy & Repair

CELLULAR HEALTH

NAD+ (Nicotinamide Adenine Dinucleotide)

The molecule at the center of cellular energy and aging research

NAD+ is a coenzyme found in every cell in your body. It plays a central role in converting food into usable energy, repairing damaged DNA, and regulating key proteins involved in aging — including sirtuins, which have been linked to longevity in research going back to the early 2000s. NAD+ levels naturally decline with age, and that decline has become one of the more studied targets in longevity medicine.

WHAT PEOPLE USE IT FOR

- Energy production and mitochondrial function
- DNA repair mechanisms
- Mental clarity and cognitive function
- Recovery from metabolic stress
- Healthy aging at the cellular level

David Sinclair's work at Harvard has kept NAD+ in the headlines for over a decade. Human trials are still catching up to animal models, but the preclinical signal is strong enough that this is one of the more defensible longevity protocols.

CELLULAR HEALTH

Glutathione

The body's master antioxidant — already inside you, but worth supporting

Glutathione is a tripeptide your body produces naturally. It's the master antioxidant — neutralizing free radicals and regenerating vitamins C and E. Levels decline with age, chronic stress, and illness. Supplementation via IV or subcutaneous delivery bypasses the poor oral bioavailability that limits most glutathione products.

WHAT PEOPLE USE IT FOR

- Neutralizing oxidative stress and free radical damage
- Liver detoxification and immune function
- Skin quality and brightness
- Cellular repair and mitochondrial health

NAD+ and Glutathione programs available at Calabasas Longevity

Physician-reviewed, compounded to order, shipped to your door.

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CHAPTER 03

Weight & Metabolism

GLP-1 / WEIGHT

Semaglutide

The most studied weight-loss compound in modern medicine

Semaglutide is a GLP-1 receptor agonist — it mimics a hormone your body naturally releases after eating that signals fullness and regulates blood sugar. The STEP 1 trial showed an average of 14.9% body weight reduction over 68 weeks. The SELECT trial — 17,600 participants — showed a 20% reduction in major cardiovascular events in people with existing CV disease and obesity.

WHAT PEOPLE USE IT FOR

- Meaningful, sustained weight loss (average ~15% in trials)
- Appetite regulation and reduced food noise
- Blood sugar stabilization and cardiovascular risk reduction

STEP 4 showed that ~two-thirds of lost weight returns within a year of stopping. This is a medication for chronic weight management, not a temporary fix. Have a long-term plan conversation with your physician before starting.

GIP/GLP-1 / WEIGHT

Tirzepatide

The next generation — activates two metabolic pathways instead of one

Tirzepatide activates both the GLP-1 and GIP receptors. The SURMOUNT-1 trial showed weight loss averaging 20.9% at the highest dose — meaningfully higher than semaglutide in head-to-head comparisons. Often used for people who've plateaued on semaglutide or who want stronger results from the start.

WHAT PEOPLE USE IT FOR

- Stronger average weight loss than GLP-1 alone
- Metabolic flexibility and improved insulin sensitivity
- Improved lipid profiles and broader metabolic outcomes

LIPOTROPIC / ENERGY

Lipo-B

A blend of lipotropic compounds (MIC: methionine, inositol, choline) plus B vitamins. Supports the liver's fat processing function and energy metabolism. Commonly paired with GLP-1 programs as a metabolic complement.

GLP-1 and metabolic programs at Calabasas Longevity

Semaglutide, Tirzepatide, and Lipo-B — physician-supervised, transparent pricing.

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CHAPTER 04

Recovery & Tissue Repair

RECOVERY / PEPTIDES

BPC-157 (Body Protection Compound 157)

One of the most studied regenerative peptides — mostly in animals, but the signal is consistent

BPC-157 is a synthetic peptide derived from a protective protein found in gastric juice. In animal models it has shown consistent effects on tendon and ligament healing, wound repair, and joint recovery. Proposed mechanisms include promotion of angiogenesis, upregulation of growth hormone receptors in tendon fibroblasts, and anti-inflammatory activity. Human clinical trials are limited — but the preclinical signal is unusually consistent across multiple research groups.

WHAT PEOPLE USE IT FOR

- Joint and tendon comfort and recovery
- Post-training repair and tissue healing
- Physical therapy support and injury recovery protocols

Honest caveat: the animal data is genuinely compelling, but the absence of large human RCTs means we're extrapolating. Source quality is critical — this compound especially requires pharmaceutical-grade preparation.

RECOVERY / PEPTIDES

BPC-157 + TB-500 (Advanced Recovery Stack)

Two regenerative peptides combined for deeper tissue repair

TB-500 (Thymosin Beta-4) is a naturally occurring peptide involved in cell migration, wound healing, and tissue regeneration. Combined with BPC-157, the two work through complementary mechanisms — BPC-157 primarily through localized tissue effects, TB-500 through systemic cell-signaling pathways.

WHAT PEOPLE USE IT FOR

- Comprehensive tendon, joint, and connective tissue support
- Recovery from heavier training loads or past injuries
- Structured recovery protocols alongside physical therapy

Recovery programs at Calabasas Longevity

BPC-157 and BPC-157 + TB-500, physician-reviewed, pre-mixed and ready to use.

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CHAPTER 05

Hormone Support

HORMONE / PEPTIDES

Sermorelin

A peptide that asks your body to make more of its own growth hormone

Sermorelin is a synthetic analog of growth hormone-releasing hormone (GHRH). Unlike synthetic HGH, which replaces the hormone directly, Sermorelin stimulates the pituitary gland to produce its own GH. The body's natural feedback loops remain intact, and GH release happens in the normal pulsatile pattern. Typically administered at night before sleep, when GH secretion is naturally highest.

WHAT PEOPLE USE IT FOR

- Restorative deep sleep (GH is primarily secreted during slow-wave sleep)
- Lean body mass support and recovery
- Energy, vitality, and cellular repair

HORMONE / PEPTIDES

Tesamorelin

More targeted growth hormone support — with FDA approval history

Tesamorelin is also a GHRH analog with a longer half-life and more targeted mechanism than Sermorelin. It was FDA-approved under the brand name Egrifta for HIV-associated lipodystrophy — making it one of the few peptides in this space with a formal FDA approval history. In off-label use, clinical data on visceral fat reduction is particularly robust.

WHAT PEOPLE USE IT FOR

- Targeted body composition and visceral fat reduction
- Lean mass preservation and long-term metabolic support
- Cognitive function (studied in older adults)

A companion lab panel is typically required before and during Tesamorelin protocols. Generally used with specific body composition goals in mind rather than as a first-line hormone support option.

Hormone support programs at Calabasas Longevity

Sermorelin and Tesamorelin, physician-supervised with companion labs.

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CHAPTER 06

Longevity & Mitochondria

LONGEVITY / MITOCHONDRIA

MOTS-C

A mitochondria-derived peptide at the intersection of energy, metabolism, and longevity

MOTS-C is a peptide encoded within mitochondrial DNA — an unusual origin for a signaling molecule, identified at USC in 2015. It acts as a mitochondrial signal that regulates nuclear gene expression, influencing metabolism, physical resilience, and cellular stress responses. In animal models, it has improved exercise capacity, metabolic function, and lifespan markers. It spans multiple categories — metabolic, longevity, and hormone-adjacent — which is why it appears in several different protocols.

WHAT PEOPLE USE IT FOR

- Mitochondrial efficiency and energy production
- Exercise tolerance and physical resilience
- Metabolic flexibility and healthy aging pathways

LONGEVITY / SLEEP

Epitalon

A tetrapeptide studied for longevity, circadian rhythm, and telomere biology

Epitalon is a synthetic tetrapeptide derived from Epithalamin, a natural pineal gland extract. Decades of research — particularly from Vladimir Khavinson's group — have studied it for telomere elongation, circadian rhythm regulation, and longevity. Studies suggest it may activate telomerase and extend telomere length. Typically used in short, twice-yearly protocols rather than continuous dosing.

WHAT PEOPLE USE IT FOR

- Telomere support and healthy aging biology
- Circadian rhythm regulation and sleep quality
- Melatonin production and neuroendocrine balance

The evidence base comes largely from Russian research, with limited independent Western replication. That said, Epitalon is one of the more studied longevity peptides in existence — the data on telomere effects is genuinely interesting even with those caveats.

MOTS-C and Epitalon at Calabasas Longevity

Longevity protocols, physician-reviewed, pre-mixed and compounded to order.

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CHAPTER 07

Cognitive Health

COGNITIVE / NOOTROPICS

Semax & Selank

Two Russian-developed peptides with a growing following in cognitive health

Semax is derived from ACTH and upregulates BDNF — a key protein in neuroplasticity — while modulating dopamine and serotonin systems. Selank is an analog of tuftsin, studied primarily for anxiolytic effects and approved as an anti-anxiety medication in Russia. Together they're often stacked for calm, focused energy: Semax for cognitive activation, Selank for stress resilience. Both are approved in Russia; neither is FDA-approved.

WHAT PEOPLE USE IT FOR

- Focus, cognitive clarity, and productivity
- Stress resilience and anxiety reduction
- Mood stability and neuroplasticity support via BDNF

Evidence comes largely from Russian research institutions. Independent Western replication is limited and long-term safety data in healthy people is thin. These are not benign supplements — physician oversight matters.

CHAPTER 08

Skin & Cellular Repair

SKIN / TISSUE

GHK-Cu (Copper Peptide)

A naturally occurring copper complex with collagen support and broad tissue repair activity

GHK-Cu is a naturally occurring copper-binding peptide that declines with age, studied since the 1970s. It promotes collagen and elastin synthesis, stimulates antioxidant enzyme production, and activates hundreds of beneficial genes while suppressing inflammation-related ones. Topically, it has an established place in dermatology. Systemically, it's used for broader tissue regeneration and anti-inflammatory effects.

WHAT PEOPLE USE IT FOR

- Skin elasticity, texture, collagen and elastin production
- Hair follicle support and wound healing
- Systemic anti-inflammatory activity and tissue repair

Cognitive and skin programs at Calabasas Longevity

Semax/Selank and GHK-Cu, physician-reviewed, pre-mixed.

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CHAPTER 09

Why Source Quality Changes Everything

Most of the conversation in the peptide space focuses on whether a compound works. That's the right question. But there's a prior question people frequently skip: *is what you're taking actually what you think it is?*

Source quality is the most underappreciated variable in this space. The risks from poor sourcing aren't hypothetical — they're well-documented.

The contamination risk most people miss

Peptides carry specific contamination risks. The main one: **endotoxins (LPS)** — fragments of gram-negative bacterial membranes that are a byproduct of manufacturing. Standard sterilization kills bacteria but doesn't destroy endotoxins. A vial can be sterile and still be loaded with endotoxins if the manufacturing environment wasn't controlled. In small amounts they cause inflammation and flu-like symptoms. In larger amounts, serious systemic reactions.

Purity ≠ Identity

A certificate of analysis showing 99% purity tells you the sample is mostly one compound. It doesn't confirm that compound is what the label says. Identity verification requires mass spectrometry — not just HPLC purity testing.

WHAT TO LOOK FOR IN A COA

- Independent third-party testing lab
- Identity via mass spectrometry
- Endotoxin testing (LAL test)
- Sterility testing for injectables

503A VS 503B COMPOUNDERS

503A — compounds per individual prescription, regulated by state pharmacy boards.

503B — FDA-registered, held to cGMP standards. Higher bar for injectables.

Red flags to watch for

⚠ No CoA available, or CoA from the same company that made the product

⚠ No endotoxin testing listed

⚠ Pricing significantly below market — quality manufacturing has real costs

⚠ No prescription required for prescription compounds

⚠ "Research use only" labeling used as a workaround for human-use compounds

⚠ Staff who can't answer specific questions about testing protocols

Pre-mixed. Compounded to order. Shipped direct.

All programs at Calabasas Longevity use licensed compounding pharmacies — no reconstitution required on your end.

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CHAPTER 10

How Physician-Supervised Care Works

If you're serious about any of the compounds in this guide, the pathway that makes the most sense — for both safety and efficacy — is physician-supervised care through a licensed compounding pharmacy. Here's what that actually looks like in practice.

1

Physician review

An independent licensed physician evaluates your health history and goals. This isn't a rubber stamp — it's a real clinical assessment of whether a protocol is appropriate for you.

2

Prescription issued

If appropriate, the physician writes a prescription for the specific compound and dosage indicated for your situation.

3

Compounding pharmacy fills it

A licensed pharmacy prepares the compound to specification. Pre-mixed formulations eliminate reconstitution steps and the handling risks that come with them.

4

Direct shipping

Compounds ship directly to you, typically refrigerated, with clear dosing instructions and physician contact for follow-up.

5

Monitoring and follow-up

A responsible protocol includes labs where indicated, check-ins on response and side effects, and dosage adjustments as needed.

What to ask before you start

- What does my health history mean for this specific compound?
- What's the dose and escalation schedule — and can we adjust if needed?
- What labs or monitoring do you recommend?
- Which compounding pharmacy do you use, and is it 503A or 503B?
- What's the long-term plan — and what happens if I stop?

Ready to explore what's right for you?

Calabasas Longevity offers physician-supervised programs across every category in this guide. Transparent pricing, licensed pharmacies, pre-mixed compounds shipped direct.

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This guide was written by Erica for Just Ask Erica (justaskerica.com). It is for educational and informational purposes only. Nothing in this guide constitutes medical advice, and no compound should be started without consulting a qualified healthcare provider. Erica is not a physician.